MICROSOFT CERTIFIED | AZ-800

STUDY NOTES

Microsoft Hyper-V

Complete Study Notes for AZ-800 Exam

COMPLETE 16 CHAPTERS

Administering Windows Server Hybrid Core Infrastructure

Covers: Virtualization | Networking | Storage | HA | Security | Azure Integration

- VM Fundamentals
- Checkpoints & Templates
- Virtual Networking
- Hyper-V Storage
- Failover Clustering
- Live Migration & Replica
- Shielded VMs & Security
- Performance & Monitoring
- PowerShell Automation
- Azure Stack HCI
- Azure Site Recovery
- Windows Admin Center
- SDN & HNV
- Storage Spaces Direct
- Troubleshooting
- Backup & DR

Windows Server 2022 | Hyper-V | Failover Clustering | Azure Stack HCI | SCVMM

Professional Study Edition | 2025

Table of Contents

Microsoft Hyper-V — AZ-800 Complete Study Notes

16 Chapters	150+ Concepts	50+ Tables	40+ PowerShell Examples	20+ Diagrams / Comparisons

01 Hyper-V Fundamentals

EXAM TIP

Hyper-V is Microsoft's enterprise hypervisor. Mastering its architecture is the foundation of the entire AZ-800 exam.

1.1 What is Hyper-V?

Microsoft Hyper-V is a Type 1 (bare-metal) hypervisor built into Windows Server and Windows 10/11 (Pro/Enterprise/Education). It creates isolated Virtual Machines (VMs) that share physical hardware resources while remaining independent of each other and the host OS.

Type 1 vs Type 2 Hypervisors

Type 1 (Bare-Metal)	Type 2 (Hosted)
Runs directly on hardware	Runs on top of a host OS
Example: Hyper-V, VMware ESXi	Example: VMware Workstation, VirtualBox
Better performance & security	Easier setup, good for desktops
Production use	Dev/Test/Personal use
Lower latency	Higher overhead

1.2 Hyper-V Architecture

Partition Model

- Root Partition (Parent) — runs Windows Server, has direct hardware access, hosts Hyper-V management stack
- Child Partitions (Guests) — run VMs, no direct hardware access, communicate through VMBus
- Hypervisor Layer — sits between hardware and partitions, enforces isolation

Key Components

- vmms.exe — Hyper-V Virtual Machine Management Service (VMMS)
- vmwp.exe — VM Worker Process (one per running VM, provides fault isolation)
- VMBus — high-speed ring-buffer communication channel between host and guest
- Integration Services — guest drivers (synthetic devices) for near-native I/O performance

WARN

If VMMS stops, running VMs continue (in their vmwp.exe processes) but NO management operations are possible until VMMS restarts.

1.3 Hardware Requirements

Requirement	Detail	Notes
64-bit Processor	Intel or AMD	Mandatory — 32-bit not supported

SLAT	Intel EPT or AMD RVI	Second Level Address Translation
Hardware Virtualization	Intel VT-x or AMD-V	Must be enabled in BIOS/UEFI
DEP	Data Execution Prevention	NX or XD bit required
RAM	Minimum 4 GB (host)	More for production workloads

KEY POINT

SLAT is mandatory from Windows Server 2012 R2 onwards. It eliminates software shadow page tables, dramatically improving VM memory management performance.

1.4 Generation 1 vs Generation 2 VMs

Generation 1 vs Generation 2	
Generation 1	Generation 2
Legacy BIOS firmware	UEFI firmware
Emulated + synthetic devices	Synthetic devices only
Broader OS compatibility	Requires 64-bit OS
IDE boot disk (emulated)	SCSI boot disk (synthetic)
No Secure Boot	Secure Boot supported
Slower boot time	Faster boot time
Windows / Linux (older)	Windows Server 2012+ / Modern Linux
No vTPM	Virtual TPM (vTPM) supported

EXAM TIP

EXAM: Generation cannot be changed after VM creation. Choose Generation 2 for all new Windows Server 2012+ and modern Linux VMs for best performance and security features.

1.5 Virtual Disk Formats

Format	Max Size	Key Feature	Use Case
VHD	2 TB	Legacy format	Older/compat scenarios
VHDX	64 TB	Journaling, 4K alignment	All new deployments
VHD Set (.vhds)	64 TB	Shared multi-VM access	Guest clustering

Virtual Disk Types

Type	Description	Best For
Fixed	Pre-allocates full size on disk	Predictable performance, production
Dynamic	Expands as data is written	Dev/test, space saving

Differencing	Stores changes relative to parent disk	Templates, checkpoints, VDI
Pass-through	Direct physical disk access (no VHD file)	Max performance, SQL Server

TIP

VHDX uses a transaction log (journal) for metadata changes. If the host crashes mid-write, VHDX recovers to a consistent state on next mount — unlike VHD which can corrupt.

02 VM Configuration & Memory Management

EXAM TIP

Memory configuration is a common exam topic. Understand Dynamic Memory, Smart Paging, and NUMA together.

2.1 Creating Virtual Machines

VMs can be created through multiple interfaces, each with different capabilities:

Method	Tool	Best For
GUI Wizard	Hyper-V Manager	Single VM, quick setup
PowerShell	New-VM cmdlet	Automation, scripting
Windows Admin Center	Browser UI	Modern management, remote
SCVMM Templates	System Center VMM	Enterprise, standardized deployments

2.2 Memory Management

Static Memory

- Fixed amount of RAM assigned at VM creation, never changes
- Recommended for: SQL Server, Domain Controllers, performance-sensitive workloads
- Predictable and simpler to capacity plan

Dynamic Memory

Hyper-V adjusts RAM allocated to a VM at runtime based on demand. Four key parameters:

Parameter	Description	Example
Startup RAM	Memory at boot	1 GB
Minimum RAM	Lowest Hyper-V can reduce to	512 MB
Maximum RAM	Upper ceiling for the VM	8 GB
Memory Buffer %	Extra headroom above demand	20%

Smart Paging

- Temporary mechanism — uses disk space as emergency memory during VM RESTART only
- Triggered when: host RAM is below VM's Minimum RAM at restart time
- NOT for sustained use — performance is very poor
- Smart paging file is placed in the VM's working directory by default

WARN

Disable Dynamic Memory for SQL Server, Domain Controllers, and memory-intensive applications. Use Static Memory with sufficient allocation for these workloads.

2.3 CPU Configuration

Setting	Purpose	Recommendation
Virtual Processors	Number of vCPUs assigned	Match to actual workload need
CPU Reserve (%)	Guaranteed CPU % for this VM	Set for critical VMs
CPU Limit (%)	Maximum CPU % cap	Use to throttle non-critical VMs
CPU Weight	Relative scheduling priority (1-10000)	Default 100; increase for priority VMs

WARN

Over-provisioning vCPUs causes CPU Ready Time to increase for ALL VMs on the host. The hypervisor must find enough free physical cores simultaneously. Fewer vCPUs often performs BETTER.

2.4 VM Checkpoints (Snapshots)

Standard vs Production Checkpoints	
Standard Checkpoint	Production Checkpoint
Saves complete VM state (memory)	No memory state saved
Captures RAM contents to .vmrs file	Uses VSS inside guest
Application-inconsistent	Application-consistent
Good for dev/test rollback	Good for production workloads
Fast to create	Slightly slower (VSS quiesce)
Default in Windows Server 2012 R2	Default from Server 2016 onwards

Checkpoint Files

- .avhdx — differencing disk storing changes since checkpoint
- .vmrs — saved memory state (Standard checkpoints only)
- .vmcx — VM configuration snapshot

WARN

Deep checkpoint chains (3-4+ levels) severely impact VM I/O performance. Every write must also update the child differencing disk. Merge checkpoints regularly in production.

Key PowerShell Commands

```
> Checkpoint-VM -VMName 'VM1' -SnapshotName 'Pre-Update'
> Restore-VMSnapshot -Name 'Pre-Update' -VMName 'VM1'
> Remove-VMSnapshot -VMName 'VM1' -Name 'Pre-Update'
> Set-VM -VMName 'VM1' -CheckpointType Production
```

03 Virtual Networking

EXAM TIP

Virtual networking questions frequently appear on AZ-800. Know all three switch types, NIC teaming, SR-IOV, RDMA, and QoS deeply.

3.1 Virtual Switch Types

Three Types of Hyper-V Virtual Switches	
External	Internal vs Private
Connects VMs to physical network	Internal: VMs + host only
Host also gets access (optional)	Private: VMs only — no host
Binds to a physical NIC	No physical NIC binding
Use for: production VM traffic	Internal: management, NAT
Requires physical switch config	Private: isolated lab testing

3.2 NIC Teaming & Switch Embedded Teaming

LBFO NIC Teaming vs SET (Switch Embedded Teaming)	
LBFO (Legacy)	SET (Modern)
OS-level teaming	Built into Hyper-V vSwitch
Not compatible with RDMA directly	Fully RDMA/SMB Direct compatible
Up to 32 NICs	Up to 8 NICs
Does not require Hyper-V	Requires Hyper-V
Being deprecated	Recommended for HCI/S2D

KEY POINT

For Azure Stack HCI and Storage Spaces Direct, always use SET (Switch Embedded Teaming), not LBFO. SET is required for RDMA-based converged networking.

3.3 Advanced Networking Features

Feature	Purpose	Requirement
SR-IOV	Near-native NIC performance via hardware VFs	SR-IOV capable NIC + switch
VMQ	Hardware-accelerated packet queuing per VM	VMQ-capable NIC
RDMA / SMB Direct	CPU-bypass high-speed transfers	RDMA NIC (iWARP/RoCE/InfiniBand)
RSS	Receive Side Scaling across CPU cores	Multi-core + RSS NIC
RSC	Receive Segment Coalescing (large receive)	RSC-capable NIC

3.4 Network Port Protection

Feature	What It Blocks	Enable On
DHCP Guard	Rogue DHCP server responses from VMs	Untrusted VM adapters
Router Guard	Router Advertisement / Redirect messages	Non-router VM adapters
Port ACLs	Inbound/Outbound traffic by rule	Per-VM adapter
MAC Spoofing	Allows VM to use multiple/different MACs	NLB, nested Hyper-V, gateways

TIP
Enable MAC address spoofing for nested virtualization (Hyper-V inside a VM) and NLB scenarios. It is disabled by default for security.

3.5 VLAN Configuration

- Access Mode — VM adapter tagged with a single VLAN ID (most common)
- Trunk Mode — VM receives multiple VLAN-tagged frames (for virtual routers/firewalls)

```
> Set-VMNetworkAdapterVlan -VMName 'VM1' -Access -VlanId 100
> Set-VMNetworkAdapterVlan -VMName 'VM1' -Trunk -AllowedVlanIdList '100-200' -NativeVlanId 1
```

3.6 Bandwidth Management (QoS)

- Minimum Bandwidth — guaranteed Mbps or weight-based minimum
- Maximum Bandwidth — cap to prevent a VM consuming all bandwidth
- Absolute Mode — specific Mbps values (predictable SLA)
- Weight-Based Mode — proportional sharing during congestion

```
> Set-VMNetworkAdapter -VMName 'VM1' -MinimumBandwidthAbsolute 100MB
-MaximumBandwidth 1GB
```

04 Hyper-V Storage

EXAM TIP

Storage configuration impacts VM performance most significantly. Know the differences between storage types, CSV, SMB 3.0, and Storage Spaces Direct.

4.1 Storage Types Overview

Storage Type	Protocol	Use Case	HA Support
Local / DAS	SATA/SAS/NVMe	Single host	No
SAN – iSCSI	TCP/IP	Shared block storage	Yes (MPIO)
SAN – Fibre Channel	FC / FCoE	High-performance SAN	Yes (MPIO)
SMB 3.0 File Share	TCP (SMB)	Scale-Out File Server	Yes
Storage Spaces Direct	Internal disks	HCI – Azure Stack HCI	Yes (S2D)

4.2 Cluster Shared Volumes (CSV)

CSV allows multiple Hyper-V cluster nodes to simultaneously read and write to the same NTFS volume, enabling seamless Live Migration without storage ownership transfer.

How CSV Works

- One node is the CSV Coordinator (handles metadata operations)
- All nodes can perform direct I/O (reads and writes) to the volume
- If coordinator fails, another node automatically takes over
- Mounted under C:\ClusterStorage\VolumeX on all nodes

KEY POINT

CSV is essential for Hyper-V failover clusters. Without CSV, VM storage ownership must transfer between nodes during migration, causing brief I/O interruption.

4.3 SMB 3.0 for Hyper-V

SMB 3.0 Feature	Benefit
SMB Direct (RDMA)	CPU-bypass transfers — dramatically lower latency
SMB Multichannel	Uses multiple NICs simultaneously — bandwidth + redundancy
Transparent Failover	Client-side reconnect — no VM storage interruption
SMB Encryption	AES encryption for data in transit
Scale-Out File Server	Active-active cluster file shares for Hyper-V

4.4 Storage Spaces Direct (S2D)

S2D pools internal disks across multiple servers to create a software-defined storage pool with built-in redundancy — the foundation of hyperconverged infrastructure (HCI).

S2D Architecture

- Storage Bus Layer (SBL) — intelligent tiering between NVMe/SSD and HDD
- Storage Spaces — software RAID across pooled disks (mirror/parity)
- CSV — exposes volumes to all Hyper-V cluster nodes simultaneously
- Supports NVMe, SSD, HDD in various tiering combinations

KEY POINT

Azure Stack HCI is built on top of Hyper-V + Storage Spaces Direct. S2D replaces the need for a dedicated SAN in hyperconverged deployments.

4.5 Storage QoS

- Minimum IOPS — guaranteed I/O floor for a VM's virtual disk
- Maximum IOPS — ceiling to prevent noisy-neighbor I/O saturation
- Policies managed via PowerShell or SCVMM
- Available for VHDX files on CSV and SMB 3.0 shares

```
> New-StorageQosPolicy -Name 'SQLPolicy' -MinimumIops 1000 -MaximumIops 5000
> Set-VMHardDiskDrive -VMName 'SQL1' -QoSPolicyID (Get-StorageQosPolicy 'SQLPolicy').PolicyId
```

4.6 ReFS Benefits for Hyper-V

ReFS Feature	Hyper-V Benefit
Block Cloning	Instantaneous VHDX creation/checkpoints — no data copy
Integrity Streams	Detects and auto-corrects data corruption on mirrors
No chkdsk required	Faster volume availability after unexpected shutdown
Large volume support	Up to 35 PB volumes — suitable for large S2D deployments

TIP

ReFS block cloning makes checkpoint creation and VM template deployment nearly instantaneous regardless of disk size, by copying metadata pointers instead of actual data blocks.

05 High Availability & Failover Clustering

EXAM TIP

HA is heavily tested on AZ-800. Master: Failover Clustering prerequisites, quorum types, Live Migration vs Quick Migration, and Hyper-V Replica.

5.1 Failover Clustering Overview

Spec	Windows Server 2019/2022
Max nodes per cluster	64
Max VMs per cluster	8,000
Max VMs per node	1,024
Quorum options	Disk, File Share, Cloud Witness
Storage options	CSV, SMB 3.0, FC SAN, iSCSI SAN

5.2 Quorum & Witness Types

Quorum Witness Options	
Disk Witness	Cloud / FS Witness
Physical shared disk	Azure Blob or File Share
Highest reliability	No extra hardware needed
SAN/iSCSI required	Ideal for 2-node or stretched clusters
Consumes a LUN	Minimal bandwidth needed (Azure)
Traditional data centers	Modern / branch office deployments

KEY POINT

Cloud Witness (Azure Blob Storage) is the recommended witness for 2-node clusters or stretched clusters across two sites. It acts as a tiebreaker without a third physical site.

5.3 VM Migration Types

Migration Type	VM Running?	Shared Storage?	Downtime	Use Case
Live Migration	Yes	Optional (Shared Nothing)	Zero	Routine moves, load balancing
Storage Live Migration	Yes	N/A (moves storage)	Zero	Move VHD to new storage
Quick Migration	No (saved)	Yes (required)	Seconds	Legacy; superseded by Live

Shared Nothing Migration	Yes	No (transfers both)	Zero	Move to different host+storage
Export / Import	No	N/A	Full downtime	Clone, archive, move offline

5.4 Live Migration Deep Dive

Prerequisites

- Both hosts in same AD domain
- Compatible processor generations
- Same virtual switch names
- Hyper-V role installed on both
- Kerberos constrained delegation OR CredSSP
- Sufficient network bandwidth
- Firewall: TCP 6600, 443
- Live Migration enabled on hosts

Live Migration Process

- 1. Pre-copy phase — VM memory pages copied to destination while VM runs
- 2. Dirty page tracking — changed pages re-copied iteratively
- 3. Blackout — very brief pause (<1ms typically) to copy final dirty pages
- 4. Switchover — CPU execution transferred to destination
- 5. VM continues running on destination; source resources freed

Key PowerShell Commands

```
> Enable-VMMigration
> Set-VMHost -VirtualMachineMigrationEnabled $true
-VirtualMachineMigrationAuthenticationType Kerberos
> Move-VM -Name 'VM1' -DestinationHost 'Host2' -IncludeStorage -DestinationStoragePath
'D:\VMs'
```

5.5 Hyper-V Replica

Hyper-V Replica asynchronously replicates VMs to a secondary host/site for disaster recovery without requiring shared storage.

Replication Interval	RPO	Notes
30 seconds	~30 sec	Windows Server 2012 R2+; lowest RPO
5 minutes	~5 min	Balanced performance/RPO
15 minutes	~15 min	Lower network bandwidth usage

Failover Types

Failover Type	Data Loss Risk	VM State	Use When
Planned Failover	Zero (full sync first)	Stops on primary	Scheduled maintenance
Unplanned Failover	Possible (last cycle)	Primary may be down	Emergency/disaster
Test Failover	None (isolated)	Replica runs in test network	DR testing/validation

Extended Replication

- Primary Replica Extended Replica (three-site DR chain)
- Extended replica configured separately with Enable-VMReplication -AsReplica

```
> Enable-VMReplication -VMName 'VM1' -ReplicaServerName 'RepHost' -AuthenticationType Kerberos -ReplicationFrequencySec 30
> Start-VMInitialReplication -VMName 'VM1'
> Get-VMReplication -VMName 'VM1'
```

06 Security Features

EXAM TIP

Shielded VMs and VBS are increasingly important security topics. Understand the full attestation flow and HGS role for the exam.

6.1 Shielded VMs Architecture

Shielded VMs protect Generation 2 VMs from unauthorized access by Hyper-V administrators, rogue hosts, or malware — even if the fabric is compromised.

Components

Component	Role
Host Guardian Service (HGS)	Attestation + key protection service
vTPM (Virtual TPM)	Enables BitLocker inside the VM
Key Protector (.p7b)	Encrypted VM key — only unlocked by HGS-approved hosts
Shielding Data File (.pdk)	Contains secrets, certificates, policies for VM provisioning
BitLocker	Encrypts all VM disks using vTPM-stored keys

Attestation Flow

- 1. Hyper-V host contacts HGS Attestation service
- 2. HGS verifies host health (TPM or Host Key method)
- 3. If approved, HGS Key Protection service releases VM's encryption keys
- 4. Hyper-V decrypts the VM disk with the released keys
- 5. VM starts — if attestation fails, VM CANNOT start on that host

TPM Attestation vs Host Key Attestation	
TPM Attestation	Host Key Attestation
Uses physical TPM 2.0 chip	Uses pre-shared public/private key pair
Strongest security	Simpler to deploy
Verifies code integrity (HVCI)	No code integrity verification
Production recommended	Lab/dev environments
Requires TPM 2.0 on host	Any hardware supported

6.2 Virtualization-Based Security (VBS)

VBS uses Hyper-V to create an isolated Virtual Secure Mode (VSM) region, protecting critical OS components from attacks even if the kernel is compromised.

VBS Feature	What It Protects	How
HVCI	Kernel code integrity	Only signed code runs in kernel mode

Credential Guard	LSASS credentials (NTLM hashes, Kerberos tickets)	Isolated in VSM — prevents PTH attacks
WDAC / Device Guard	Application control policy	Blocks unsigned executables
Secure Kernel	Hyper-V kernel itself	Runs in higher privilege than OS kernel

KEY POINT

Credential Guard uses Hyper-V VBS to store credential material in an isolated container. Even if an attacker gains SYSTEM privileges on the host OS, they cannot access credentials protected by Credential Guard.

6.3 Secure Boot & vTPM

Secure Boot Templates (Generation 2)

Template	Guest OS
Microsoft Windows	Windows Server / Windows 10+
Microsoft UEFI Certificate Authority	Linux using shim bootloader (Ubuntu, RHEL, SUSE)
Open Source Shielded VM	Linux Shielded VMs (without shim)
No Secure Boot	Legacy/other OS that lacks Secure Boot support

WARN

Wrong Secure Boot template = VM boot failure. Always match template to guest OS. For Linux VMs, use 'Microsoft UEFI Certificate Authority' unless otherwise specified.

6.4 Secured-core Server (Windows Server 2022)

Layer	Technology	Protection
Hardware	TPM 2.0 + DRTM	Boot firmware integrity measurement
Firmware	Secure MOR + Kernel DMA Protection	Prevents DMA attacks
OS	VBS + HVCI + Credential Guard	Kernel and credential isolation
Virtualization	Hyper-V Hypervisor	Enforces all isolation boundaries

07 Performance Monitoring & Optimization

EXAM TIP

Know the key performance counters, Dynamic Memory behavior, and common bottleneck identification techniques for the exam.

7.1 Key Performance Counters

Counter	What It Measures	Alert When
Hyper-V Hypervisor LP - % Guest Run Time	Time running guest code	High is good (efficient)
Hyper-V Hypervisor LP - % Hypervisor Run Time	Hypervisor overhead	Consistently > 10% = concern
Hyper-V Hypervisor LP - % Root Run Time	Host OS overhead	High = host CPU contention
Hyper-V VP - CPU Wait Time per Dispatch	vCPU waiting for pCPU	> 1ms = CPU overcommit
Hyper-V VM Vid Partition - Physical Pages Allocated	Guest memory pages in host RAM	Dropping = memory pressure
Hyper-V Virtual Storage Device - Queue Length	Pending I/O requests	Sustained > 2 = I/O bottleneck
Hyper-V Virtual Storage Device - R/W Bytes/sec	Storage throughput	Compare to storage capacity

7.2 Memory Optimization

Memory Ballooning

- Occurs when host is under memory pressure — guest balloon driver inflates
- Hyper-V reclaims memory by inflating the balloon driver inside the guest
- Guest OS thinks it has less memory — may start paging to disk
- Solution: Add host RAM, reduce VM density, or tune Dynamic Memory minimums

NUMA Optimization

- NUMA = Non-Uniform Memory Access — some RAM is faster for a given CPU
- Keep VM vCPUs and memory within a single NUMA node for best performance
- NUMA spanning allows VM to exceed single node — causes remote memory latency
- Configure vNUMA topology to match physical NUMA for NUMA-aware apps (SQL Server)

TIP

For SQL Server VMs: disable Dynamic Memory, configure vNUMA to match physical NUMA topology, and size the VM to fit within a single NUMA node if possible.

7.3 Storage Performance

Issue	Cause	Solution
High Queue Length	I/O bottleneck on storage	Storage QoS, faster storage tier
High disk latency	Too many dynamic expanding VHDXs	Use fixed-size VHDX
Slow checkpoint operations	VHD format or deep chain	Use VHDX on ReFS, merge chains
Antivirus impact	AV scanning VHDX files on host	Exclude Hyper-V directories from AV
I/O congestion	Noisy-neighbor VM	Apply Storage QoS IOPS limits

7.4 Resource Metering

Resource Metering tracks per-VM resource usage over time for chargeback, capacity planning, and reporting.

```
> Enable-VMResourceMetering -VMName 'VM1'
> Measure-VM -Name 'VM1' # Returns CPU, memory, disk, network stats
> Reset-VMResourceMetering -VMName 'VM1'
```

Hyper-V Power Plan

- Use 'High Performance' power plan on Hyper-V hosts
- 'Balanced' plan throttles CPU frequency during low load, causing VM latency spikes

```
> powercfg -setactive SCHEME_MIN # Sets High Performance
```

08 PowerShell Automation

EXAM TIP

PowerShell is integral to AZ-800. Be able to write and interpret cmdlets for VM management, networking, storage, and migration.

8.1 Essential Cmdlet Reference

Cmdlet	Purpose
New-VM	Create a new virtual machine
Get-VM	List VMs and their state
Start-VM / Stop-VM	Start or stop a VM (Stop-VM -Force = hard off)
Set-VM	Modify VM settings (memory, checkpoints, etc.)
Suspend-VM / Resume-VM	Pause / resume a VM (keeps in RAM)
Save-VM	Save VM state to disk (frees RAM)
Remove-VM	Delete a VM (add -Force to skip prompt)
Export-VM / Import-VM	Export to folder / Import from folder
Checkpoint-VM	Create a checkpoint (snapshot)
Restore-VMSnapshot	Revert VM to a checkpoint
Move-VM	Live Migration to another host
Enable-VMReplication	Configure Hyper-V Replica

8.2 VM Creation & Configuration

```
> # Create Generation 2 VM with 4GB RAM and 60GB dynamic VHDX
> New-VM -Name 'WebServer01' -Generation 2 -MemoryStartupBytes 4GB \
> -NewVHDPath 'D:\VMs\WebServer01.vhdx' -NewVHDSizeBytes 60GB \
> -SwitchName 'ExternalSwitch'
```

```
> # Set 4 vCPUs and enable Dynamic Memory
> Set-VMProcessor 'WebServer01' -Count 4
> Set-VMMemory 'WebServer01' -DynamicMemoryEnabled $true -MinimumBytes 1GB
-MaximumBytes 8GB
```

```
> # Enable nested virtualization
> Set-VMProcessor 'WebServer01' -ExposeVirtualizationExtensions $true
> Set-VMNetworkAdapter 'WebServer01' -MacAddressSpoofing On
```

8.3 Networking Cmdlets

```
> # Create virtual switches
> New-VMSwitch -Name 'ExtSwitch' -NetAdapterName 'Ethernet' -AllowManagementOS $true
> New-VMSwitch -Name 'IntSwitch' -SwitchType Internal
```

```
> New-VMSwitch -Name 'PrivSwitch' -SwitchType Private
```

```
> # Configure VLAN and network adapter
> Set-VMNetworkAdapterVlan -VMName 'VM1' -Access -VlanId 100
> Set-VMNetworkAdapter -VMName 'VM1' -DhcpGuard On -RouterGuard On
```

8.4 Storage Cmdlets

```
> # Create and manage virtual disks
> New-VHD -Path 'D:\VMs\Data.vhdx' -SizeBytes 200GB -Fixed
> New-VHD -Path 'D:\VMs\Data.vhdx' -SizeBytes 200GB -Dynamic
> Resize-VHD -Path 'D:\VMs\Data.vhdx' -SizeBytes 300GB
> Optimize-VHD -Path 'D:\VMs\Data.vhdx' -Mode Full # Compact
> Test-VHD -Path 'D:\VMs\Data.vhdx' # Check integrity
```

```
> # Attach/detach disks
> Add-VMHardDiskDrive -VMName 'VM1' -Path 'D:\VMs\Data.vhdx'
> Remove-VMHardDiskDrive -VMName 'VM1' -ControllerType SCSI -ControllerNumber 0
-ControllerLocation 1
```

8.5 PowerShell Direct

PowerShell Direct executes commands inside a VM from the Hyper-V host without using the network stack — works even if VM networking is broken.

```
> # Run command inside VM without network
> Invoke-Command -VMName 'VM1' -ScriptBlock { Get-Service -Name W32Time }
> # Enter interactive session
> Enter-PSSession -VMName 'VM1' -Credential (Get-Credential)
```

09 Azure Integration & Hybrid Cloud

EXAM TIP

Azure integration is a major AZ-800 topic. Understand Azure Stack HCI, Azure Arc, Azure Site Recovery, and Azure Migrate thoroughly.

9.1 Azure Stack HCI

Azure Stack HCI vs Windows Server + Hyper-V

Azure Stack HCI	Windows Server + Hyper-V
Delivered as Azure cloud service	Traditional on-premises product
Requires Azure subscription	No Azure subscription needed
Continuous monthly updates	Periodic Windows Update
Integrated Azure management	Optional Azure integration
HCI validated hardware catalog	Any server hardware
Azure Hybrid Benefit pricing	Standard licensing
Built-in S2D + Hyper-V + SDN	Components installed separately

KEY POINT

Azure Stack HCI is the recommended platform for NEW hyperconverged deployments. It provides the tightest Azure integration and most current Hyper-V features.

9.2 Azure Arc for Hyper-V VMs

Azure Arc projects on-premises Hyper-V guest VMs into Azure, enabling Azure management services without moving workloads to the cloud.

Services Enabled by Azure Arc

- Azure Policy — compliance enforcement
- Azure Monitor — metrics and logs
- Microsoft Defender for Cloud
- Azure Update Manager — patch management
- Change Tracking and Inventory
- Azure Automation DSC
- Microsoft Sentinel integration
- Azure Automanage — auto-enrollment

Arc Setup Steps

- 1. Download and install the Connected Machine agent on the Hyper-V guest
- 2. Agent registers the machine with Azure Resource Manager
- 3. Machine appears in Azure portal under Azure Arc > Servers
- 4. Assign Azure policies, enable Defender, configure monitoring

9.3 Azure Site Recovery (ASR) for Hyper-V

Hyper-V Replica vs Azure Site Recovery	
Hyper-V Replica	Azure Site Recovery
Secondary on-premises site needed	Azure as recovery site — no secondary site
Max 30-second RPO	Near-continuous replication to Azure
Free (built into Windows Server)	Paid Azure service
Manual failover orchestration	Automated recovery plans
No test failover isolation automation	One-click test failover with isolation
No runbook integration	Azure Automation runbook integration

ASR Setup for Hyper-V

- 1. Create Recovery Services Vault in Azure
- 2. Add Hyper-V site (can be standalone or SCVMM-managed)
- 3. Install Azure Site Recovery Provider on Hyper-V hosts
- 4. Enable replication per VM — select target Azure region, storage, network
- 5. Run Test Failover to validate DR before real failover

9.4 Azure Migrate for Hyper-V

Phase	Tool	Output
Discovery	Azure Migrate appliance deployed on-premises	Inventory of VMs, apps, dependencies
Assessment	Azure Migrate: Assessment	Azure VM sizing, cost estimates, readiness
Migration	Azure Migrate: Migration	Replicate VMs to Azure (agentless for Hyper-V)

TIP

Azure Migrate uses Hyper-V Replication protocol for agentless VM replication. No agent is needed inside the guest OS — the appliance interacts directly with Hyper-V on the host.

9.5 Windows Admin Center (WAC)

WAC Capability	Description
VM Management	Create, configure, start/stop, console access
Cluster Management	Failover Cluster and Azure Stack HCI management
Storage Management	Storage Spaces Direct, CSV, disk management
Network Management	Virtual switch, SDN configuration
Azure Integration	Arc registration, ASR, Azure Monitor, Azure Backup
Extensions	Third-party extensions for additional functionality

10 Advanced Features

EXAM TIP

Advanced features like Nested Virtualization, DDA, and GPU-P are increasingly tested. Know when to use each and their requirements.

10.1 Nested Virtualization

Nested Virtualization exposes hardware virtualization extensions to a guest VM, allowing Hyper-V to run inside a VM. Used for dev/test labs, training, and containerization.

Requirements & Setup

Step	Command / Action
VM must be OFF	Stop-VM -Name 'LabVM'
Enable virt extensions	Set-VMProcessor 'LabVM' -ExposeVirtualizationExtensions $true
Enable MAC spoofing	Set-VMNetworkAdapter 'LabVM' -MacAddressSpoofing On
Disable Dynamic Memory	Set-VMMemory 'LabVM' -DynamicMemoryEnabled $false
Start VM	Start-VM -Name 'LabVM'

WARN

Nested virtualization is NOT recommended for production workloads — additional hypervisor overhead and security considerations apply. Use for dev/test only.

10.2 Discrete Device Assignment (DDA)

DDA passes a physical PCIe device (GPU, NVMe, FPGA) directly to a VM, bypassing the Hyper-V virtualization layer for near-native hardware performance.

Aspect	Detail
Device types	GPU (for rendering/compute), NVMe SSDs, FPGA, specialized NICs
Performance	Near-native — bypasses hypervisor I/O stack
Isolation	Device assigned exclusively to ONE VM only
Snapshot support	Not supported while DDA device is assigned
VM constraint	VM must support DDA; device drivers required inside guest

10.3 GPU Partitioning (GPU-P) — Windows Server 2022

GPU-P divides a single physical GPU into up to 16 virtual GPU partitions, each assigned to a separate VM — enabling GPU resource sharing unlike DDA (which is exclusive).

DDA vs GPU Partitioning	
DDA (Discrete Device Assignment)	GPU-P (GPU Partitioning)
Entire GPU to ONE VM	GPU shared across up to 16 VMs
Maximum GPU performance	Shared GPU performance
No snapshot/checkpoint	Supports checkpoints
Any PCIe device	GPU-specific feature
For GPU-intensive single workloads	VDI, AI inference, mixed workloads

10.4 Software Defined Networking (SDN)

SDN Component	Function
Network Controller	Central REST API management of virtual/physical network
Software Load Balancer (SLB)	North-south and east-west load balancing for VMs
RAS Gateway	BGP routing, VPN, and NAT for tenant networks
HNV / VFP	VXLAN/NVGRE overlay networks for tenant isolation
SDN Express	PowerShell deployment automation for SDN stack

10.5 VM Groups

- VM Groups allow batch operations on multiple VMs simultaneously
- Management Group — contains VMs directly
- Collection Group — contains Management Groups (hierarchical)

```
> New-VMGroup -Name 'AppTier' -GroupType VMCollectionType
> Add-VMGroupMember -VMGroupName 'AppTier' -VM (Get-VM 'Web1','Web2','Web3')
> Checkpoint-VMGroup -Name 'AppTier' -SnapshotName 'PreDeploy'
```

11 Troubleshooting Guide

EXAM TIP

Troubleshooting scenarios are common exam questions. Use a systematic approach: check event logs, verify config, test connectivity, check resources.

11.1 Troubleshooting Framework

Step	Action	Tools
1. Identify	Gather symptoms, error messages, timeline	Event Viewer, Hyper-V logs
2. Check resources	CPU, memory, storage, network	Task Manager, perfmon, Get-VM
3. Verify config	VM settings, switches, VHD paths	Hyper-V Manager, PowerShell
4. Check logs	VMMS-Admin, VMMS-Operational, System	Event Viewer (Microsoft-Windows-Hyper-V-*)
5. Test connectivity	Network, storage, cluster	ping, Test-NetConnection, Cluster Validator
6. Apply fix	Based on findings	PowerShell, GUI, registry

11.2 Common Issues & Solutions

Issue	Root Cause	Solution
VM fails to start	Insufficient host RAM / locked VHD	Check resources, verify VHD access
Live Migration fails	Kerberos delegation not configured	Configure constrained delegation / CredSSP
VM loses network	Wrong vSwitch or VLAN mismatch	Verify adapter settings with Get-VMNetworkAdapter

Replica not syncing	Firewall blocking TCP 80/443	Allow replication ports on firewall
High CPU Ready	Too many vCPUs (overcommit)	Reduce vCPU count, migrate VMs
Storage queue depth high	Noisy neighbor or slow storage	Apply Storage QoS, upgrade storage
Checkpoint merge slow	Dynamic VHDX on NTFS	Move to ReFS, use fixed VHDX
Clock drift in VMs	Time sync misconfiguration	Configure w32tm properly, disable host sync for DCs

11.3 Key Hyper-V Event Logs

Event Log	Path	Contains
VMMS-Admin	Microsoft-Windows-Hyper-V-VMMS-Admin	Critical errors, VM failures
VMMS-Operational	Microsoft-Windows-Hyper-V-VMMS-Operational	VM lifecycle events
Replication-Admin	Microsoft-Windows-Hyper-V-VMMS-Replication	Replica errors/status
Worker-Admin	Microsoft-Windows-Hyper-V-Worker-Admin	Per-VM runtime errors
Live Migration	Microsoft-Windows-Hyper-V-VMMS-Migration	Migration errors/events

11.4 Cluster Troubleshooting

Split-Brain Scenario

Occurs when cluster nodes lose communication but both believe they own cluster resources. Quorum witness prevents this by requiring a majority vote.

Node Eviction Troubleshooting

- Check System event log on evicted node for errors before eviction
- Verify cluster heartbeat network connectivity (ping cluster network IPs)
- Check storage accessibility (CSV availability) from evicted node
- Review Cluster events in Failover Cluster Manager

```
> Test-Cluster -Node 'Node1','Node2' -Include 'Storage','Network' # Validate cluster
> Get-ClusterNode # View node status
> Get-ClusterResource | Where-Object State -ne 'Online' # Find offline resources
```

12 Backup & Disaster Recovery

EXAM TIP

Understand the difference between Hyper-V Replica (DR), Azure Site Recovery (cloud DR), and backup solutions. Know RTO/RPO concepts for exam.

12.1 Backup Solutions Overview

Solution	Type	RPO	Best For
Windows Server Backup	Built-in VSS backup	Hours	Small environments, free
System Center DPM	Enterprise backup	Minutes-Hours	Large organizations
Azure Backup (MABS)	Cloud-integrated backup	Hours	Azure-connected hybrid
Veeam / Commvault	Third-party	Minutes	Enterprise feature-rich
Hyper-V Replica	Async replication (not backup)	30 sec-15 min	DR — NOT backup
Azure Site Recovery	Cloud DR replication	Near-continuous	Cloud-based DR

WARN

Hyper-V Replica is a DR (disaster recovery) tool, NOT a backup solution. It does not protect against accidental deletion or data corruption. Use both Replica AND a backup solution.

12.2 VSS (Volume Shadow Copy Service)

VSS coordinates application-consistent backups by quiescing (freezing) application writes before taking a snapshot, then resuming writes.

VSS Component	Role
VSS Requestor	Backup application that initiates the backup (WSB, DPM, Veeam)
VSS Provider	Creates/manages the snapshot (OS or hardware-based)
VSS Writer	Application-side component that quiesces data (SQL Writer, Exchange Writer)
Hyper-V VSS Writer	Coordinates backup of VMs from the host — uses Integration Services

12.3 RTO & RPO Guidance

DR Technology	Typical RPO	Typical RTO	Notes
Hyper-V Replica (30s)	30 seconds	Minutes	Manual failover; no SLA automation
Azure Site Recovery	Near-zero	Minutes-1hr	Automated failover plans available

Failover Clustering	Near-zero (sync)	Seconds-Minutes	HA — not DR across sites
Stretched Cluster + Storage Replica	Near-zero (sync)	Seconds	Active-active sites
Azure Backup + ASR	Hours (backup) + mins (ASR)	Hours + Minutes	Combined backup+DR

12.4 DR Testing Best Practices

- Perform Test Failover quarterly minimum — use isolated test network
- Verify application functionality, not just VM boot, during test
- Document RTO achieved during tests vs RTO objective
- Update runbooks based on test findings
- Test recovery from backup separately from Replica test failover

```
> Start-VMFailover -VMName 'VM1' -AsTest # Test failover (isolated network)
> Stop-VMFailover -VMName 'VM1' # Clean up test failover
> Start-VMFailover -VMName 'VM1' # Real unplanned failover
> Complete-VMFailover -VMName 'VM1' # Commit planned failover
```

13 Installation & Role Configuration

EXAM TIP

Know all installation methods, configuration settings, and delegation options — including Server Core and remote management setup.

13.1 Installing the Hyper-V Role

Method	Command / Steps
Server Manager (GUI)	Add Roles and Features > Server Roles > Hyper-V > Reboot
PowerShell (GUI server)	Install-WindowsFeature -Name Hyper-V -IncludeManagementTools -Restart
PowerShell (Server Core)	Install-WindowsFeature -Name Hyper-V -IncludeManagementTools
DISM	dism /Online /Enable-Feature /FeatureName:Microsoft-Hyper-V /All

13.2 Hyper-V Host Configuration (Set-VMHost)

Setting	PowerShell Parameter	Notes
Default VM path	VirtualMachinePath	Where .vmcx files are stored
Default VHD path	VirtualHardDiskPath	Where VHD/VHDX files are stored
Live Migration	VirtualMachineMigrationEnabled $true	Enable/disable globally
Concurrent migrations	MaximumVirtualMachineMigrations	Default: 2 in/out
Migration auth type	VirtualMachineMigrationAuthentication Type	Kerberos or CredSSP
NUMA spanning	NumaSpanningEnabled $false	Disable for performance
Enhanced Session Mode	EnableEnhancedSessionMode $true	RDP console features

13.3 Remote Management

Setup Steps for Remote Hyper-V Management

- 1. Enable WinRM: winrm quickconfig on the Hyper-V host
- 2. Ensure DNS resolution works between management PC and host
- 3. Configure Windows Firewall rules (ports 135, 445, 5985/5986)
- 4. Add host in Hyper-V Manager: Connect to Server > Enter hostname
- 5. For workgroup: configure CredSSP or certificate-based auth

Remote Management Tools

Tool	Protocol	Best For
Hyper-V Manager (MMC)	WinRM / DCOM	GUI management of 1-10 hosts
Windows Admin Center	HTTPS / WinRM	Modern browser-based, cluster support

PowerShell Remoting	WinRM	Automation and scripting
SCVMM Console	SCVMM Agent	Enterprise — 100+ hosts

13.4 Delegation & Access Control

Access Level	Method	Grants
Full Hyper-V admin	Local Hyper-V Administrators group	All VM operations, no host admin
Full server admin	Local Administrators group	Everything including host config
Enterprise admin	SCVMM User Roles	Granular RBAC — VM Admin, Fabric Admin, etc.
Live Migration	Kerberos Constrained Delegation	VMMS cross-host authentication

14 Failover Clustering Deep Dive

EXAM TIP

Clustering is a major exam domain. Master all quorum configurations, CAU, stretched clustering, and Storage Replica.

14.1 Cluster Requirements & Validation

Prerequisites

- All nodes in same AD domain
- Same Windows Server version (recommended)
- Shared storage (CSV, SAN, SMB)
- Redundant network paths
- Consistent hardware configuration
- Validated with Cluster Validation Wizard
- DNS registration for all NICs
- WinRM enabled for management

WARN

ALWAYS run cluster validation before creating a production cluster. Microsoft support requires a clean validation report for supportability. Use: Test-Cluster -Node 'N1','N2'

14.2 Quorum Configuration

Config	Votes	Use When
Node Majority	Each node = 1 vote	Odd number of nodes
Node + Disk Witness	Nodes + 1 disk	Even number of nodes with shared storage
Node + File Share Witness	Nodes + 1 file share	Even nodes, branch office
Node + Cloud Witness	Nodes + Azure blob	Even nodes, no third site available
No Majority (Disk Only)	1 disk	Not recommended — single point of failure

14.3 Cluster-Aware Updating (CAU)

CAU automates the rolling update process for Hyper-V cluster nodes, maintaining VM availability during the entire patching cycle.

CAU Process (Per Node)

- 1. Drain — Live Migrate all VMs off the target node
- 2. Pause — Suspend node from accepting new VM workloads
- 3. Update — Install Windows Updates
- 4. Reboot — If required by updates
- 5. Resume — Node rejoins cluster, VMs can migrate back
- 6. Repeat — Next node in the cluster

CAU Modes

Self-Updating vs Remote-Updating	
Self-Updating Mode	Remote-Updating Mode
CAU runs on the cluster itself	CAU runs from a remote orchestrator
No external management server needed	Useful for monitoring and scheduling
Simple setup	More control over scheduling
Good for most deployments	Enterprise environments with SCOM/SCVMM

14.4 Stretched Clustering & Storage Replica

Stretched clustering spans cluster nodes across two physical sites for site-level disaster recovery with automatic VM failover.

Component	Role	Replication
Storage Replica	Block-level volume replication	Synchronous (zero data loss) or Async
Cloud Witness	Quorum tiebreaker	Azure Blob (no third site needed)
Hyper-V Failover Cluster	Automatic VM failover between sites	N/A — uses S.R. storage
Fault Domain Awareness	Distribute VMs across sites	Chassis/Rack/Site topology

KEY POINT

Storage Replica is synchronous within ~5ms RTT between sites. For sites farther apart, use asynchronous replication mode (small RPO). Synchronous guarantees zero data loss.

15 SCVMM & Enterprise Management

EXAM TIP
SCVMM is relevant for enterprise AZ-800 scenarios. Know templates, dynamic optimization, fabric management, and self-service concepts.

15.1 SCVMM Architecture

Component	Function
VMM Server	Central management engine — stores all configuration in SQL DB
VMM Console	GUI management client — connect to VMM Server
VMM Agent	Installed on managed Hyper-V hosts — executes operations
VMM Library	Stores VHDs, ISOs, templates, profiles, scripts
SQL Server	VMM database — stores all fabric, VM, and policy configuration
Self-Service Portal	Web UI for tenant VM self-service (via WAP integration)

15.2 VM Templates & Profiles

Object	Contains	Purpose
Hardware Profile	vCPU, RAM, boot order, NIC, disks	Standardize VM hardware config
Guest OS Profile	Hostname, domain join, admin password, timezone	Automate OS customization
VM Template	Hardware Profile + OS Profile + base VHD	One-click VM deployment
Service Template	Multi-tier application with multiple VMs	Deploy complex apps as one unit

15.3 Dynamic Optimization & Power Optimization

Dynamic vs Power Optimization	
Dynamic Optimization	**Power Optimization**
Balances VM load across cluster nodes	Consolidates VMs onto fewer hosts
Uses Live Migration	Uses Live Migration + Wake-on-LAN / IPMI
Triggered by CPU/memory thresholds	Triggered by low overall demand period
Goal: prevent resource contention	Goal: reduce energy consumption
Runs continuously/scheduled	Runs during off-peak hours

15.4 Logical Networking in SCVMM

Logical Networks abstract physical VLANs into named network resources that VMs can consume by name, decoupling VM configuration from physical network topology.

Object	Description
Logical Network	Named abstraction of physical network (e.g., 'Production Network')
Network Site	VLAN + IP subnet mapping within a logical network
IP Pool	Static IP address pool for automated IP assignment
VM Network	Tenant-facing network built on a Logical Network
Port Profile	Virtual switch port settings (bandwidth, VLANs)

16 AZ-800 Exam Preparation & Quick Reference

EXAM TIP

This chapter consolidates the most exam-critical facts, common gotchas, and PowerShell cheat sheet for final revision.

16.1 AZ-800 Exam Skills — Hyper-V Domains

Skill Domain	Weight	Key Topics
Deploy & Manage AD DS	25-30%	AD, RODC, Sites, Replication
Manage Windows Server & Workloads Hybrid	10-15%	Azure Arc, Update Mgmt, WAC
Manage VMs & Containers	15-20%	Hyper-V, Docker, AKS-HCI, Checkpoints
Implement On-Premises & Hybrid Networking	15-20%	vSwitch, SDN, DNS, DHCP, NPS
Manage Storage & File Services	15-20%	S2D, CSV, SMB, DFS, Azure Files

16.2 Critical Facts to Remember

KEY POINT

These facts are commonly tested. Memorize them before your exam.

Fact	Value / Detail
Max cluster nodes (Server 2022)	64 nodes
Max VMs per cluster	8,000
Max VMs per node	1,024
Max RAM per VM (Server 2022)	240 TB
Max vCPUs per VM (Server 2022)	2,048
Max VHDX size	64 TB
Hyper-V Replica intervals	30 seconds, 5 minutes, 15 minutes
Default checkpoint type (Server 2016+)	Production Checkpoint (VSS-based)
Default Live Migration auth	Kerberos Constrained Delegation
Live Migration TCP port	6600 (+ 443 for Kerberos)
Generation 1 max vCPUs	64
Generation 2 max vCPUs (2022)	2,048
SLAT required from	Windows Server 2012 R2 onwards
SET max NICs	8 physical adapters

S2D max servers	16 (Azure Stack HCI)

16.3 Common Exam Gotchas

WARN
Generation CANNOT be changed after VM creation — choose carefully upfront.

WARN
Hyper-V Replica is NOT a backup — it does not protect against accidental deletion. Use backup software additionally.

WARN
Disable Dynamic Memory for SQL Server, DCs, and Exchange VMs. Use Static Memory with sufficient allocation.

WARN
LBFO NIC teaming is deprecated in favour of SET for Hyper-V. Use SET for Storage Spaces Direct and RDMA-based converged networking.

WARN
VM Configuration Version upgrade is ONE-WAY — it cannot be rolled back. The VM becomes incompatible with older Hyper-V host versions.

WARN
Disable host time sync (Integration Services > Time Synchronization) for Domain Controllers. DCs should sync from the domain time hierarchy, not the Hyper-V host.

16.4 Ultimate PowerShell Quick Reference

Task	Cmdlet
Create a Generation 2 VM	New-VM -Name 'VM1' -Generation 2 -MemoryStartupBytes 4GB -NewVHDPath 'C:\VM1.vhdx' -NewVHDSizeBytes 80GB -SwitchName 'vSwitch'
Enable Dynamic Memory	Set-VMMemory 'VM1' -DynamicMemoryEnabled $true -MinimumBytes 1GB -MaximumBytes 16GB
Set vCPUs	Set-VMProcessor 'VM1' -Count 4
Create external vSwitch	New-VMSwitch 'ExtSwitch' -NetAdapterName 'Ethernet' -AllowManagementOS $true
Enable Live Migration	Enable-VMMigration; Set-VMHost -VirtualMachineMigrationEnabled $true
Live Migrate VM	Move-VM 'VM1' -DestinationHost 'Host2' -IncludeStorage -DestinationStoragePath 'D:\VMs'
Create checkpoint	Checkpoint-VM 'VM1' -SnapshotName 'Pre-Update' -SnapshotType Production
Enable Hyper-V Replica	Enable-VMReplication 'VM1' -ReplicaServerName 'RepHost' -AuthenticationType Kerberos -ReplicationFrequencySec 30
Check replica health	Get-VMReplication -VMName 'VM1'
Enable nested virtualization	Set-VMProcessor 'VM1' -ExposeVirtualizationExtensions $true

Resize VHDX	Resize-VHD -Path 'C:\VM1.vhdx' -SizeBytes 200GB
Get all running VMs	Get-VM | Where-Object State -eq 'Running'
Enable resource metering	Enable-VMResourceMetering 'VM1'; Measure-VM 'VM1'
Run command in VM (no network)	Invoke-Command -VMName 'VM1' -ScriptBlock { hostname }

16.5 Study Checklist

- [ ] Hyper-V architecture & partitions
- [ ] VM generations and differences
- [ ] Dynamic Memory & Smart Paging
- [ ] All 3 virtual switch types
- [ ] NIC Teaming vs SET
- [ ] SR-IOV, RDMA, VMQ concepts
- [ ] VHD vs VHDX vs VHD Set
- [ ] CSV fundamentals
- [ ] S2D & HCI architecture
- [ ] Failover Clustering prereqs
- [ ] Live Migration full process
- [ ] Hyper-V Replica & all failover types
- [ ] Shielded VMs & HGS attestation
- [ ] Azure Arc & ASR configuration
- [ ] CAU rolling update process
- [ ] PowerShell cmdlets hands-on

TIP

Set up a Hyper-V lab (physical or nested in Azure). Hands-on practice with PowerShell and the GUI is the single most effective preparation for AZ-800.